

3rd Grade Truss Bridge

(Coach Scheduled Event - 45 Minutes)



Description: Teams will design and construct a truss bridge that spans a 30.0 cm gap between two supports. The bridge will be tested by supporting a 300 g mass and, if successful, a 500 g mass for 30 seconds each. Teams will also complete a written assessment covering bridge types, structural design, and forces acting within bridges.

Participants per assigned Team Number: 2

- If your school has 1 team you will send 2 students
- If your school has 2 teams you will send 2 students per team number; team numbers may not intermix
- If your school has 3 teams you will send 2 students per team number; team numbers may not intermix

Safety Equipment - Teams Must Bring:

- **Safety glasses labeled ANSI Z87+ (impact rated)**
- **All competitors must wear their eye protection during the entire event.**
- **If a team does not have the required eye protection, they will be given the opportunity to obtain it, time allowing, but will not receive extra time.**
- **If a team is unable to obtain eye protection, the team will not compete and will receive a no-show score.**

Impact Safety Glasses Required



Eyeglasses are not safety rated!

Teams must also bring:

- Pencil, pen, or marker
- Ruler or straightedge
- A pair of scissors per team member

Materials Provided at Event:

- 30 paper drinking straws approximately 20 cm (8 in) each
- 1 meter (36 in) of 2.54 cm (1 in.) tape
- Two support blocks cut from commercial 2 x 4 lumber, of approximately 30 cm (11.8 in) long
- One wooden loading block approximately 5 cm x 5 cm x 2 cm (2 in x 2 in x 3/4 in)
- One 300 g (11 oz) mass
- One 500 g (18 oz) mass

The Competition:

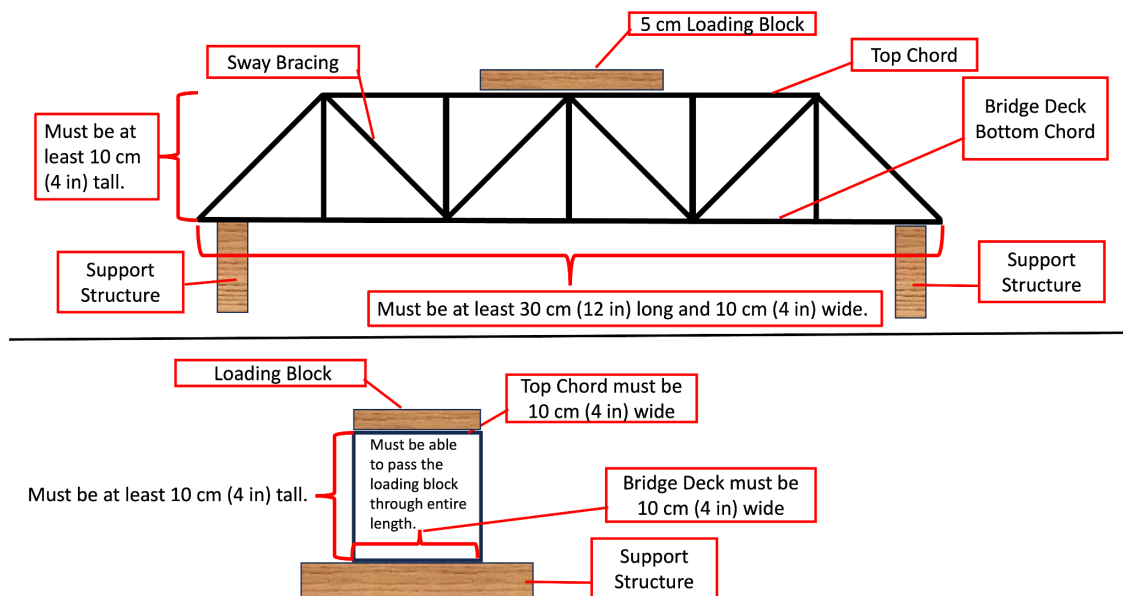
Construction Phase (30 minutes):

The bridge must:

- Span a 30.0 cm gap between supports.

- Be constructed entirely from the provided straws and tape. Teams may cut, bend, or crimp the straws and may slide them inside one another. Teams may not bring any preconstructed components or use materials not provided by the Event Supervisor.
- Have a minimum width of 10.0 cm at its midpoint, measured from outside edge to outside edge.
- Be at least 10 cm tall at its midpoint. Height is measured from the top surface of the supports to the top of the top chord.
- Allow the square face of the loading block to pass completely through the bridge along its entire length without obstruction and without turning or twisting the loading block.
- Support the loading block applied at the center apex of the bridge.

Competition Diagram (picture not drawn to scale):



The Written Assessment: Simultaneously with the device testing time, the team will take a written assessment. Teams will be called up from their written assessment to test their device and then will return to their written assessment while other teams test their device. The assessment topics may include:

- Types of bridges
- Bridge design principles
- Structural forces (tension, compression, shear, torsion)
- Load distribution
- Failure modes
- Engineering applications of trusses

Device Testing Setup:

- The Event Supervisor (ES) will place the support blocks on a level surface with the inner edges 30.0 cm (12 in) apart.
- Teams will place their bridge on the supports. No sticky surface of the tape may contact the tops or sides of the supports. If the team's bridge does not span the distance, the supports will be moved closer together for testing and the team will receive a percentage penalty (see Scoring).

Device Testing Measurements

- Before testing, the team will demonstrate that the loading block passes completely through the bridge.
- Before testing, the ES will record:
 - Whether the bridge spans the gap.
 - Whether the width requirement of 10 cm is met.
 - Whether the height requirement of 10 cm is met.
 - The height of the bridge at its center to the nearest 0.1 centimeter.
 - The number of unused, complete straws.

Load Testing

- The team will place the loading block at the center of the bridge.
- Timing begins once the 300 g mass is placed on the loading block and is fully released from the team's hand.
- Once testing begins, competitors may not touch the bridge. A bridge successfully supports a load if it remains intact and all parts of the bridge remain above the top level of the supports for 30 seconds.
- If the bridge successfully supports the 300 g mass for 30 seconds, the procedure is repeated using the 500 g mass.
- Testing ends when:
 - Any part of the bridge falls below the top level of the testing surface
 - The load falls off the loading block
 - The bridge is touched by competitors after testing begins.
 - Thirty seconds have elapsed.
 - Students remove their protective eyewear.

Scoring:

$$\text{Final Score (FS)} = \text{Assessment Score} + \text{Bonus} \times \text{Bridge Score}$$

- Bridge Score = $(\text{THS} \div \text{CHS}) \times 60$, maximum is 60 pts
 - THS (Team Height Score) = Penalties x TH
 - TH (Team Height): Height of the team's bridge at the center
 - CHS (Competition Height): Maximum THS over all the teams
- Assessment Score = $(\text{TAS} \div \text{CAS}) \times 40$, maximum is 40 pts
 - TAS (Team Assessment Score): Team written assessment score
 - CAS (Competition Assessment Score): Highest written assessment score across teams
- Penalties -
 - .70 multiplier (30% penalty)
 - Bridge does not hold the 300g weight for 30 seconds
 - .20 multiplier (80% penalty)
 - Bridge width is less than 10cm at the midpoint
 - Loading block does not pass through the bridge
 - Tape is used to stick the bridge to the support structure
 - The height of the bridge is less than 10cm at the midpoint.
 - Both penalties can be applied.
- Bonus
 - If the bridge can hold the 500 g mass for 30 seconds, the Bridge will receive a 20% bonus.

Scoring Example

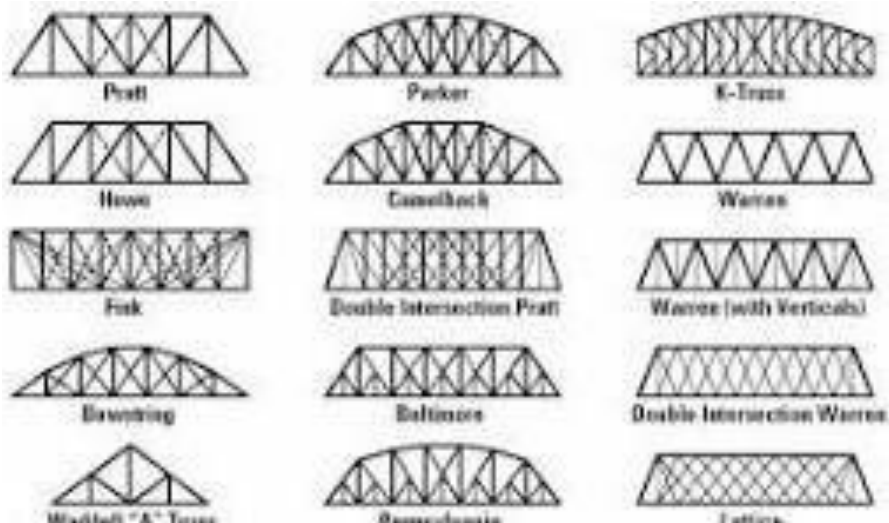
- Competition Assessment Score (CAS) = 9.
- The team scores 6 on the assessment (TAS).
 - Assessment Score for the team= $(6 \div 9) \times 40 = 26.67$ points
- Team's height = 12 cm, there are no penalties. The bridge also held the 500 g weight.
- The CHS = 15 cm
- Bridge Score = $(12 \div 15) \times 60 = 48$ points
- Final Score (FS) = $26.67 + 1.2 \times 48 = 84.27$

Tiebreakers:

- Number of unused straws
- Select test questions

Resources:

- <https://www.youtube.com/watch?v=oVOnRPefcno>
- <https://thestemlaboratory.com/straw-bridges/>
- <https://www.stem-inventions.com/truss-bridge> (This site uses materials that are different from the one used for this event.)
- Science Teachers & Librarians
- Get permission from your parents/coach to use AI to create a study guide using the rules provided here
- Pictures of Truss Bridges:



Allowed Calculator

This is the acceptable calculators for all Division A events.

Class I - Stand-alone non-graphing, non-programmable, non-scientific 4-function or 5-function calculators. These are the most basic types of calculators. They are limited to the four basic mathematical functions and square roots. They are often cheap, costing as low as a dollar online.



Safety Glasses

Indirect Vent Goggles or Safety glasses are a mandatory safety requirement for some events. It is important to choose the type of protection specific to the task. Competitors are responsible for providing their own protective eyewear. Teams will not be allowed to compete without adequate eye protection.

Category C: Indirect Vent Goggles



Category B: Safety glasses



Compliant Eyewear Categories:

CATEGORY B

- Description: Impact protection. They provide protection from a high inertia particle hazard (high mass or velocity)
- Corresponding ANSI designation/required marking: Z87+.
- Category B eyewear can look either like goggles or safety glasses, but must have the Z87+ marking embossed on them.

CATEGORY C

- Description: Indirect vent chemical/splash protection goggles. These seal tightly to the face completely around the eyes and have indirect vents constructed so that liquids do not have a direct path into the eye (or no vents at all). If you can see through the vent holes from one side to the other, they are NOT indirect vents
- Corresponding ANSI designation/required marking: Z87 (followed by D3 is the most modern designation but, it is not a requirement)

Examples of Non-Compliant Eyewear:

- Face shields/visors are secondary protective devices and are not approved.
- Prescription Glasses containing safety glass should not be confused with safety spectacles. "Safety glass" indicates the glass is made to minimize shattering when it breaks. Unless these glasses bear the Z87+ mark they are not approved for use.

Notes:

1. A goggle that bears the Z87+ mark and is an indirect vent chemical/splash protection goggle will qualify for both Category B & C
2. VisorGogs do not seal completely to the face, but are acceptable as Type C goggles